

**MULTIMEDIA UNIVERSITY OF KENYA
FACULTY OF COMPUTING AND IT**

Department of Computer Science

**BSc SOFTWARE ENGINEERING – Y2S2
CSE 2222 DATA COMMUNICATIONS AND COMPUTER NETWORKS**

**ASSIGNMENT
MAY 2026**

This assignment is part of the continuous assessment for the course unit that contributes 30% of the total mark. You are expected to work individually.

Note: Plagiarism will be checked for your submission

SUBMISSION: This assignment is to be typed and submitted in soft copy through the MMU eLearning platform (Moodle) on or before **23:55 hrs. Friday 5th June 2026**. No email or late submissions will be accepted.

TOTAL: 70 MARKS

Answer all the questions.

Question 1 (10 Marks)

You are the administrator of a LAN that has increased the number of users by a factor of three over the past couple of years. You have been adding users by increasing the number of switches and switch ports on the network. Now users are complaining that network response is slowing and even their computers seem slower. You monitor the network for a few days and find that broadcast frames constitute a high percentage of your network traffic.

- a. What can you do to contain the broadcasts? (4 marks)
- b. What network reconfiguration might you have to do? Justify your decision. (6 Marks)

Question 2 (10 marks)

You must install 125 computers for a new business that wants to run TCP/IP and have access to the Internet. The ISP in town will assign you only four public IP addresses, so you decide to assign the computers addresses in the range 172.16.1.1/16 through 172.16.1.125/16.

What else must you do to allow these computers to access the Internet? Explain your reasons.

Question 3 (50 marks)

You are the team leader contracted to design and install a dedicated communication link between the capital city and county offices, which are 250 km apart. There is no existing link, thus starting afresh. The client instructs that the project must be within budget, and be able to sustain a real-time AI-enabled business application. Further, the county office will also serve as a disaster recovery site for the primary server which is in the capital city.

Answer the following questions, **justifying your answer**.

- a. What would you recommend for the following? (8 marks)
 - i. Transmission medium
 - ii. Transmission mode
 - iii. Bandwidth
 - iv. Network device(s)
- b. For your answers in (a) above, list two disadvantages for each item. (8 marks)
- c. Which communication protocols will be deployed? (4 marks)
- d. Design the addressing that will ensure safe delivery of data through the link. (10 marks)
- e. Which measures will be deployed to ensure secure data transmission? (10 Marks)
- f. Draw a well labelled diagram depicting your design as answered above. (10 marks)

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